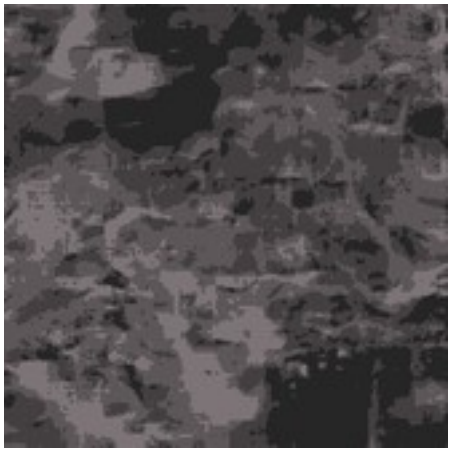
SAR Image Analysis Report

Original SAR (not found; preview shown)

Colorized



K-Means Segmented

Metadata:

- Original File: Not available
- Colorized File: colorized_1764150106.png
- Processed On: 2025-11-26 15:12:20 IST
- SAR Classifier Confidence: 0.000
- Colorization Model: Pix2Pix GAN
- KMeans Clusters: 4
- Colorized Resolution: 256 x 256 px

Cluster Analysis:

Cluster 0: Mixed / Other Surface | Mean RGB: (68, 64, 66) | Area: 33.9%

Cluster 1: Likely Soil / Barren Land (Reddish Tone) | Mean RGB: (124, 116, 118) | Area: 13.68%

Cluster 2: Mixed / Other Surface | Mean RGB: (93, 87, 90) | Area: 32.87%

Cluster 3: Mixed / Other Surface | Mean RGB: (39, 38, 39) | Area: 19.54%

Land Cover Summary:

- Mixed / Other Surface → 33.9%
- Likely Soil / Barren Land (Reddish Tone) → 13.68%
- Mixed / Other Surface → 32.87%
- Mixed / Other Surface → 19.54%

Note: Colorized image is GAN-generated and for visualization only.

Analysis Details & Recommendations

Interpretation Notes:

The area is dominated by mixed / other surface (86.31% combined).

Secondary presence: likely soil / barren land (reddish tone) (13.68%).

K-Means groups pixels by color similarity — labels are heuristic.

Consider adding multispectral or time-series data for improved analysis.

Recommendations:

The analysis shows dominant land cover is mixed/other (34.04%). The area contains mixed surfaces; further targeted analysis (e.g., higher-resolution imagery) is recommended to refine land-cover details. Secondary clusters show distributed variations; using time-series imagery would help confirm seasonal changes. Recommended actions: collect ground-truth samples where possible, prioritize monitoring based on dominant class, and re-run analysis after incorporating higher-resolution data or additional bands.

Confidence Indicators:

SAR Detection Confidence: 0.000
Dominant Class Percentage: 33.90%
Classification Reliability: High

Limitations:

- KMeans is unsupervised and may not match exact land-cover classes.
- GAN colorization is synthetic and does not represent true spectral values.
- Classification depends on image quality and may contain uncertainty.
- Single-image analysis; use multi-temporal data for trend-based conclusions.

Final Summary:

The analysis indicates predominant vegetation with mixed surfaces.

No major water bodies or dense urban zones detected in this image.

Recommend periodic monitoring and ground-truth sampling to improve confidence.